

§ 微分方程.

代数方程: $P_n(x) = 0$.

$n \geq 5$ 没有求根公式.

超越方程, $\begin{cases} e^x \\ \sin x / \cos x \end{cases} \quad \begin{cases} \sin x = x \\ x e^x = 1 \end{cases}$

微分方程: 已知函数及其导数/微分方程

$0 \leq \eta \rightarrow n \leq \eta$
 $\downarrow \quad \quad \downarrow$
 $f \quad \quad f^{(n)}(x)$

设 $f(x)$ 在 $I = (a, b)$ 上连续的, $y(x)$ 为 $f(x)$ 原函数.

$$y'(x) = f(x).$$

$$\underbrace{\frac{dy}{dx}}_{y(x)} = \underbrace{f(x)}_{\text{已知}}$$

$\overline{\mathbb{R}}$
 任何 $\in \mathbb{R}$ 上 $y(x)$

$$y = y(x)$$

$$\left(\frac{dy}{dx} \right)$$

随 x 变.

$$\underline{x^2 + y^2 = 25.}$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0.$$

$$2x + 2y \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

$$\frac{dy}{dx} = f(x, y)$$

解的存在性 } 存在.
 唯一性 }
 连续 }
 光滑 }

- 一阶 (常) 微分方程.

y(x),

$$\underline{\underline{F = ma}}$$

$$F = F(t)$$

$$a = \frac{dv}{dt}$$

$$F(t) = m \cdot \frac{dv}{dt}$$

$$\boxed{\frac{dv}{dt}} = \frac{1}{m} F(t) \quad \text{--- 一阶}$$

$$v = \frac{ds}{dt}$$

$$F(t) = m \frac{d^2s}{dt^2}$$

$$\underline{\underline{\frac{d^2s}{dt^2}}} = \underline{\underline{\frac{1}{m} F(t)}} \quad \text{--- 二阶}$$

例: 一个物体以初速度 v_0 作自由落体, 受到空气阻力和重力 v 时,

g.

$$\underline{\underline{m \frac{dv}{dt} = mg - kv}}$$

$$v(t) = \frac{mg}{k} (1 - e^{-\frac{k}{m}t})$$

$$v(0) = v_0$$

$$\frac{ds}{dt} = v(t)$$

$$s(0) = 0$$

$$s(t) = \frac{mg}{k} + (v_0 - \frac{mg}{k}) e^{-\frac{k}{m}t}$$

$$m \frac{dv}{dt} = mg e^{-\frac{k}{m}t}$$

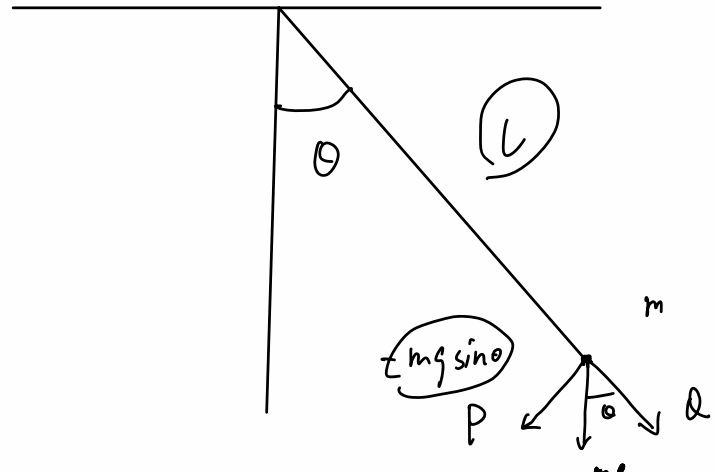
$$= mg - kv$$

$$= mg - k \left(\frac{mg}{k} + (v_0 - \frac{mg}{k}) e^{-\frac{k}{m}t} \right)$$

$$= mg e^{-\frac{k}{m}t}$$

例: 无阻尼单摆

$$\ddot{\theta} = \alpha = \frac{d^2\theta}{dt^2}$$



$$M = -mg \sin \theta = Z \alpha$$

$$= m l^2 \ddot{\theta}$$

$$l \ddot{\theta} = -g \sin \theta$$

$$\frac{d^2 \theta}{dt^2} + \frac{g}{l} \sin \theta = 0. \quad (\text{非线性})$$

$$\theta \leq 5^\circ. \quad \sin \theta \approx \theta.$$

$$\frac{d^2 \theta}{dt^2} + \frac{g}{l} \theta = 0$$

$$\theta(t) = A \sin\left(\sqrt{\frac{g}{l}} t + \varphi\right)$$

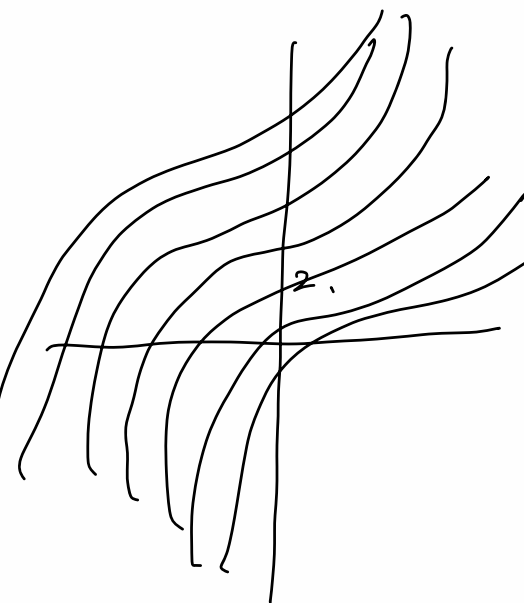
例: $f(x, y)$ 令 $3x^2$, 过点 $(1, -1)$ 方程.

$$\begin{cases} \frac{dy}{dx} = 3x^2 \\ y(1) = -1 \end{cases}$$

$$y(x) = \int 3x^2 dx = \underline{\underline{x^3 + C}}$$

$$x=1 \quad y=-1$$

$$-1 = 1 + C$$



$$C = -2.$$

$$y = x^3 - 2.$$

$$y = \boxed{f(x)}$$

一元(常)微分方程. 迄今未知函数 的导数/微分 的方程.

$$\underline{y' = xy} \quad - \text{例1}$$

方程的解.

$$\underline{x^3 \boxed{y'''} + x^2 y'' - 4xy' = 3x^2}$$

二阶方程.

$$\underline{(t^2 + x)dt + xdx = 0} \quad - \text{例1}$$

$$- \text{例1} : \quad \underline{y' = f(x, y)} \quad \underline{y' - f(x, y) = 0}$$

$$F(x, y, y') = 0.$$

$$n \text{ 阶} : \quad \underline{F(x, y, y', \dots, \boxed{y^{(n)}})} = 0.$$

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$$

$$\boxed{y^{(n)}} = \boxed{C}$$

$$y = \varphi(x), \quad (\varphi \text{ 又在 } \mathbb{R} \text{ 上 } \dot{\varphi} \neq 0)$$

$$F(x, y(x), y'(x), \dots, y^{(n)}(x)) = 0$$

①通解: 解集合任意非空, 且任意非空子集和微分方程相符.

$$\frac{dy}{dx} = y.$$

$$y = C e^x$$

$$\frac{d^2 y}{dx^2} + y = 0.$$

$$y = C_1 \sin x + C_2 \cos x$$

$$\frac{d^2 y}{dx^2} = -y$$

↓
没有 x, y'

$$y' = f(x, y)$$

$$y|_{x_0} = y_0.$$

②特解: 定解条件 { 初值.
边界条件.

$$\left\{ \begin{array}{l} y'' = f(x, y, y') \\ y|_{x_0} = y_0, \quad y'|_{x_0} = y_0' \end{array} \right.$$

一元方程: 特解. 积分曲线.

①. 通解不存在:

$$(y')^2 + 1 = 0. \quad \text{无解.}$$

$$(y')^2 + y^2 = 0. \quad y=0 \text{ 特解.}$$

②. 通解不一定存在.

$$(x+y)y' = 0.$$

$$\begin{cases} y' = 0 & \rightarrow y = C. & \text{通解} \\ x+y=0 & \rightarrow y = -x & \text{特解} \end{cases}$$

求解微分方程.

一阶微分方程解法.

$$F(x, y, y') = 0 \quad \text{或} \quad y' = f(x, y)$$

$$p(x, y) dx + q(x, y) dy = 0$$

$$\int p(x, y) dx + \int C dy = 1$$

§ 线性
非线性

$$y' + p(x)y = q(x)$$

$$y'' + p(x)y' + q(x)y = f(x)$$

$$\begin{cases} x(y')^2 - 2yy' + x = 0 \\ yy' - 2xy = 3 \\ y' - \cos y = 1 \end{cases}$$

§ 一阶线性:

$$y' + p(x)y = q(x)$$

设 $p(x)$ 和 $q(x)$ 在区间 $I = (a, b)$ 上连续.

$$\begin{cases} q(x) \equiv 0 & \text{线性齐次方程} \\ q(x) \text{ 不恒为 } 0 & \text{线性非齐次方程} \end{cases}$$

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

齐次方程

齐次方程: k 次齐次方程.

$$f(ax) = a^k f(x)$$

1. 线性变换

$$y = ax$$

线性变换



$$y = ax + b \quad \text{仿射变换}$$

$$b \neq 0$$

②. 仿射变换.

$$x^5 + 2x^3y^2 + 9xy^4$$

$$\begin{array}{l} x^5 \left(1 + 2 \left(\frac{y}{x} \right)^2 + 9 \left(\frac{y}{x} \right)^4 \right) \\ y^5 \left(1 + \left(\frac{y}{x} \right)^4 + 6 \left(\frac{y}{x} \right)^7 \right) \end{array}$$

仿射: $\frac{dy}{dx} + p(x)y = 0$

$$\frac{dy}{y} = -p(x) dx$$

$$\int \frac{dy}{y} = - \int p(x) dx$$

$$\ln y = \underline{\underline{- \int p(x) dx}} + C.$$

$$y = \underline{\underline{C e^{-\int p(x) dx}}}} \quad \text{通解.}$$

$$\underline{\underline{C}} \cdot \underline{\underline{v.}}$$

例1: 求 $f(x)$. 设 $f(x) = \int_0^{2x} f\left(\frac{t}{2}\right) dt + \ln 2$, 求 $f(x)$.

解: $f'(x) = f(x) \cdot 2$. 令 $y = f(x)$.

$$y' = 2y. \quad y' - 2y = 0.$$

$$p(x) = -2.$$

$$f(x) = y = C e^{\int -2 dx} = \underline{\underline{C e^{2x}}}$$

$$f(0) = \ln 2 = C.$$

特解 $f(x) = \ln 2 \cdot e^{2x}.$

§ 非齐次.

$$\frac{dy}{dx} + p(x)y = q(x), \quad (1)$$

$$p(x), \frac{p}{2}, p(x), \dot{p} - \dot{q} \text{ 等}.$$

$$\text{同} \quad e^{P(x)}$$

$$\underbrace{e^{P(x)} \frac{dy}{dx} + \underbrace{e^{P(x)} p(x)}_{(e^{P(x)})' y}} = e^{P(x)} q(x),$$

$$(e^{P(x)} y)' = e^{P(x)} q(x),$$

$$\Rightarrow \underbrace{e^{P(x)} y = \int e^{P(x)} q(x) dx + C.}$$

$$\Rightarrow y = e^{-P(x)} \left(\int e^{P(x)} q(x) dx + C \right)$$

通解:

$$= e^{-\int p(x) dx} \left(\int e^{\int p(x) dx} q(x) dx + C \right) \\ = \boxed{C e^{-P(x)}} + e^{-P(x)} \int e^{P(x)} q(x) dx$$

齐次方程通解

非齐次方程特解

$$\boxed{a x} + \boxed{b}$$

$$y' + p(x)y = q(x),$$

$$y_1, y_2.$$

$$(y_1, -y_2) \text{ 是 } y' + p(x)y = 0 \text{ 的基.}$$

非齐次

例1: $y' + \frac{y}{x} = \frac{\sin x}{x}$ 通解

$$\begin{array}{ccc} y' + \frac{1}{x}y & = & \frac{\sin x}{x} \\ \downarrow & & \downarrow \\ p(x) & & q(x) \end{array}$$

$$\int \frac{1}{x} dx = \ln|x|$$

$$y = e^{-\ln|x|} \left(\int e^{\ln|x|} \frac{\sin x}{x} dx + C \right)$$

$$= \frac{1}{x} \left(\int \frac{|x|}{x} \cdot \frac{\sinh x}{x} dx + C \right)$$

$x > 0$
 $x < 0$

$$= \frac{1}{x} \left(\int x \cdot \frac{\sinh x}{x} dx + C \right)$$

$$= \frac{1}{x} (-\cos x + C). \quad \text{已解.}$$

例: $\frac{1}{x} \cdot (1+y^2) y dx + 2(2xy^2-1) dy = 0 \quad \text{② 已}$

以 x 为自变量
 以 y 为因变量

$$\frac{dy}{dx}$$

$$(1+y^2)y + 2(2xy^2-1) \frac{dy}{dx} = 0.$$

已解

$$(1+y^2)y \frac{dx}{dy} + 2(2xy^2-1) = 0.$$

$$\underline{(1+y^2)y} \frac{dx}{dy} + \frac{4xy^2}{(1+y^2)y} = 2.$$

$$\underline{\underline{\frac{dx}{dy}}} + \frac{4y}{1+y^2} x = \frac{2}{\underline{(1+y^2)y}}$$

$\downarrow$ $p(y)$
 $\downarrow$ $q(y)$

$$P(y) = \int p(y) dy = 4 \int \frac{y}{1+y^2} dy = 2 \int \frac{(1+y^2)'}{1+y^2} dy = 2 \ln(1+y^2)$$

$$x(y) = \frac{e^{-2 \ln(1+y^2)}}{e^{-\ln(1+y^2)^2}} \left(\int e^{2 \ln(1+y^2)} \cdot \frac{2}{(1+y^2)y} dy + C \right)$$

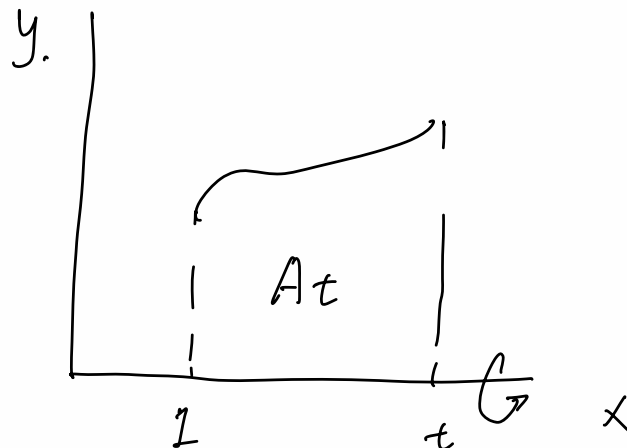
$$= \frac{1}{(1+y^2)^2} \left(\int (1+y^2)^2 \cdot \frac{2}{\cancel{(1+y^2)}} \cdot \frac{1}{y} dy + C \right)$$

$$= \frac{1}{(1+y^2)^2} \left(\int \frac{2(1+y^2)}{y} dy + C \right)$$

$$= \frac{1}{(1+y^2)^2} (2 \ln y + y^2 + C)$$

例 7.2.3. 设 $f(x) \geq 0$, $f(x) > 0$.

$$\underline{V_x = \pi t A_t}$$



解: $A_t = \int_1^t f(x) dx$

$$V_x = \int_1^t \pi (f(x))^2 dx$$

$$\int_1^t \pi (f(x))^2 dx = \pi t \int_1^t f(x) dx$$

证:

$$(f(t))^2 = \int_1^t f(x) dx + t f(t)$$

当 $t=1$

$$(f(1))^2$$

求导.

$$2f(t) f'(t) = f(t) + f(t) + t f'(t)$$

$$= 2f(t) + t f'(t)$$

$$f(1) = 0$$

$$f(1) = 1$$

$$\text{令 } y = f'(t)$$

$$2y y' = 2y + t y'$$

$$(2y - t) y' = 2y \quad \begin{matrix} \text{当 } t=1 \\ y=1 \end{matrix}$$

$$\frac{dy}{dx} = y' = \frac{2y}{2y-t} \quad \text{求 } y' \text{ 的表达式}$$

$$\frac{dt}{dy} = \frac{2y-t}{2y} = 1 - \frac{1}{2y} t$$

$$\frac{dt}{dy} + \frac{1}{2y} t = 1$$

$\downarrow$ $\downarrow$
 $p(y)$ $q(y)$

$$p(x) = \int p(y) dy$$

$$= \frac{1}{2} \ln y$$

$$+ = e^{-\frac{1}{2} \ln y} \left[\int e^{\frac{1}{2} \ln y} 1 dy + C \right]$$

$$= \frac{1}{\sqrt{y}} \left(\int \sqrt{y} dx + C \right)$$

$$= \frac{1}{\sqrt{y}} \left(\frac{2}{3} y^{\frac{3}{2}} + C \right) = \frac{2}{3} y + \frac{C}{\sqrt{y}}$$

$$t=1 \quad y=1$$

$$1 = \frac{2}{3} + C \Rightarrow C = \frac{1}{3}$$

$$\underline{x = \frac{1}{3} \left(\frac{1}{\sqrt{y}} + 2y \right)}$$

(2): $y' = \frac{\cos y}{\cos y \sinh - x \cosh y}$ (2) (3)

$\frac{dy}{dx}$

$$\frac{dx}{dy} = \sinh 2y - \tanh y x$$

$$\frac{dx}{dy} + \tanh y x = \sinh 2y \quad \int \tanh y dy$$

$\frac{dy}{dx}$ $\downarrow$
 $p(x)$ $\downarrow$
 $q(x)$

$$= \boxed{-\ln |\cos y|} + C.$$

$$x = e^{\ln |\cos y|} \left(\int e^{-\ln |\cos y|} \sin y \, dy + C \right)$$

$$= \cos y \left(\int \frac{1}{\cos y} 2 \sin y \cos y \, dy + C \right)$$

$$= \cos y (C - 2 \cos y).$$

§ 伯努利方程.

$$\begin{array}{l} \downarrow \\ y' + p(x)y = q(x)y^\alpha \quad (\alpha \neq 0, 1) \\ \hline y=0 \text{ 平凡解} \end{array}$$

约去 y^α 后.

$$y^{-\alpha} y' + p(x) y^{1-\alpha} = q(x)$$



$$\frac{1}{1-\alpha} (y^{1-\alpha})' + p(x) y^{1-\alpha} = q(x)$$

$$\text{令 } z = y^{1-\alpha}$$

$$z' + (1-\alpha)p(x)z = \underbrace{(1-\alpha)q(x)}_{\text{记为 } f(x)}$$

$$\underline{\underline{y^{1-\alpha} = z = 0 \left(\int + C \right)}}$$

解:

$$y' = \frac{y}{x(x^2y-1)}$$

$y=0$ 平凡解.

$$\frac{dx}{dy} = \frac{x(x^2y-1)}{y} = x^3 - \frac{x}{y}$$

$$\frac{dx}{dy} + \frac{1}{y}x = x^3$$

$$\downarrow$$

$$p(y)$$

$$q(y)$$

$$\alpha = 3.$$

$$z = x^{1-\alpha} = x^{-2}$$

$$z' - \frac{2}{y} z = -2$$

$$\downarrow$$

$$p(y) = -\frac{2}{y}$$

$$p(y) = -2 \ln y$$

$$\downarrow$$

$$q(y) = -2$$

$$z = \underline{e^{2 \ln y}} \left(\int e^{-2 \ln y} \cdot (-2) dy + C \right)$$

$$= y^2 \left(\int \frac{1}{y^2} (-2) dy + C \right)$$

$$= y^2 \left(\frac{2}{y} + C \right) = \underline{2y + Cy^2}$$

$$\text{let } z = x^{-2}$$

$$\frac{1}{x^2} = 2y + cy^2$$

$$x^2 (2y + cy^2) = 1$$

解:

$$\frac{dy}{dx} - \frac{y}{1+x} = xy^2 \quad \text{伯努利方程.}$$

$$\alpha = 2.$$

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{1+x} \frac{1}{y} = x$$

$$\frac{d(-\frac{1}{y})}{dx} + \frac{1}{1+x} (-\frac{1}{y}) = x$$

$$\frac{d(\frac{1}{y})}{dx} + \frac{1}{1+x} \frac{1}{y} = -x$$

$$(1+x) \frac{d(\frac{1}{y})}{dx} + \frac{1}{y} = -x(1+x)$$

$$\downarrow \quad (1+x) \cdot \frac{1}{y}$$

$$\frac{d\left(\frac{1+x}{y}\right)}{dx} = -x(1+x)$$

$$\frac{1+x}{y} = -\frac{1}{2}x^2 - \frac{1}{3}x^3 + C.$$

$$y = -\frac{6(1+x)}{2x^3 + 3x^2 + C} \quad \text{通解.}$$

$\approx$

一阶线性:

$$y' + p(x)y = q(x)$$

$e^{\int p(x)dx}$

$$q(x) = 0 \quad \text{齐次}$$

$$q(x) \neq 0 \quad \text{非齐次.}$$

$$P(x) = \int p(x)dx$$

$$y = e^{-p(x)} \left[\int e^{p(x)} q(x) dx + C \right]$$

=

$$C e^{-p(x)}$$

一阶非线性: 伯努利

$$y' + p(x)y = q(x) \underline{y^\alpha} \quad (\alpha \neq 0, 1)$$

$$\underline{y^{-\alpha} y' + p(x) y^{1-\alpha} = q(x)}$$

$$\text{令 } z = y^{1-\alpha}$$

$$z' + (1-\alpha)p(x)z = (1-\alpha)q(x)$$

§ 可分离变量的微分方程.

$$\frac{dy}{dx} = g(x)h(y)$$

或

$$\underline{M(y) dy = N(x) dx}$$

$$dM(y), dN(x),$$

$$\frac{1}{h(y)} dy = g(x) dx$$

若 $y = \varphi(x)$ 是它的解.

$g(x), h(y), M(y), N(x)$ 都在某个区间 I 上连续.

$$M(\varphi(x)) d\varphi(x) = N(x) dx$$

||

$$\int M(\varphi(x)) \varphi'(x) dx = \int N(x) dx$$

$$\int M(y) dy = \int N(x) dx$$

$G(y)$ 与 $M(y)$ 在 $[-c, b]$ 上连续

$F(x) \dots N(x)$ 在 $[-c, b]$ 上连续.

$$G(y) = F(x) + C$$

$$y = \varphi(x, C)$$

例: 7.2.2.

$$\underline{\underline{\frac{dy}{dx} = \frac{1}{x+y}}}$$

解一:

$$\frac{dx}{dy} = x+y.$$

线性方程

$$\begin{array}{ccc} \frac{dx}{dy} - x = y. & & \\ \downarrow & & \downarrow \\ p(y) = -1 & & q(y) = y. \end{array}$$

$$x = C e^y - y - 1$$

解二:

$$\Delta u = x+y.$$

$$\underline{\underline{\frac{dy}{dx} = \frac{1}{u}}}$$

$$\underline{\underline{\frac{du}{dx} = 1 + \frac{dy}{dx}}} = 1 + \frac{1}{u} = \underline{\underline{\frac{u+1}{u}}}$$

$$\frac{du}{dx} = \underline{\underline{\frac{u+1}{u}}}$$

$$\frac{u}{u+1} du = dx$$

$$\text{代入 } u=x+y.$$

$$u - \ln(u+1) = x + C$$

$$(x+y) - \ln(x+y+1) = x + C$$

$$\ln(x+y+1) = y + C$$

$$x+y+1 = ce^y$$

$$\underline{x = ce^y - y - 1}$$

12.1): $\frac{dy}{dx} = \frac{1}{x \sin^2(xy)} - \frac{y}{x}$ (可解)

解:

$$\hat{z} = xy.$$

$$\frac{dz}{dx} = y + x \frac{dy}{dx}$$

$$\begin{aligned} \frac{dz}{dx} &= y + x \left(\frac{1}{x \sin^2(z)} - \frac{y}{x} \right) \\ &= \frac{1}{\sin^2 z} \end{aligned}$$

$$\sin^2 z \, dz = dx$$

$$2z - \sin 2z = 4x + C$$

$$\text{代 } z = xy.$$

$$2xy - \sin(2xy) = 4x + C$$

可解

例: 7.17. 细菌的繁殖速度为 k . 初始细菌数为 M_0 .

设时刻 t , 细菌数为 $M(t)$

$$\begin{cases} \frac{dM(t)}{dt} = -k M(t) \\ M(0) = M_0 \end{cases}$$

$$\frac{1}{M} dM = -k dt$$

$$\ln M = -kt + C.$$

$$M = C e^{-kt}$$

$$t=0 \quad M = M_0 \quad C = M_0.$$

$$M = M_0 e^{-kt}$$

$$\frac{k \cdot 2}{1600}$$

例: 求 $\oint f(x,y) y dx + g(x,y) x dy = 0$ 的通解.

解: 令 $u = xy$ $du = x dy + y dx$

$$f(u) y dx + g(u) \underline{x dy} = 0 \quad \left(du - y dx \right)$$

$$\underline{f(u)} y dx + g(u) \underline{(du - y dx)} = 0.$$

$$(f(u) - g(u)) y dx + g(u) du = 0.$$

$$g(u) du = \underline{(g(u) - f(u)) y dx}$$

$$u = xy$$

$$y = \frac{u}{x}$$

$$\Leftrightarrow \frac{g(u)}{u(g(u) - f(u))} du = \frac{1}{x} dx$$

$$\ln(x) = \int \dots du + C.$$

$$\underline{f(tx) = t^k f(x)},$$

$k \neq \frac{1}{x}$
 $\mathbb{R}$

④ 齐次微分方程.

$$\boxed{\frac{dy}{dx}} = \frac{\varphi(x, y)}{f(\frac{y}{x})}$$

$y = tx$

齐次微分方程.

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$\uparrow u = \frac{y}{x}$

$$\text{令 } u = \frac{y}{x} \quad \underline{y = ux}$$

$$\frac{dy}{dx} = x \frac{du}{dx} + u = f(u)$$

$$x \frac{du}{dx} = f(u) - u$$

$$\frac{1}{f(u) - u} du = \frac{1}{x} dx$$

合起来

例 7.2.8.

$$x^2 y' + xy = y^2 \quad \text{求过点 } (1, 1) \text{ 的解}$$

解:

$$\underline{x^2 y' = y^2 - xy}$$

$$\underline{y' = \left(\frac{y}{x}\right)^2 - \frac{y}{x}} \quad \text{齐次}$$

$$\text{令 } u = \frac{y}{x} \quad \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$x \frac{du}{dx} = u^2 - 2u$$

$$\frac{du}{u(u-2)} = \frac{1}{x} dx$$

$$\int \frac{1}{u(u-2)} du$$

$$= \frac{1}{2} \int \left(\frac{1}{u-2} - \frac{1}{u} \right) du$$

$$= \frac{1}{2} \ln \frac{u-2}{u}$$

$$\ln \frac{u-2}{u} = 2 \ln x + C$$

$$\frac{u-2}{u} = Cx^2 \quad \text{代} \quad u = \frac{y}{x} \quad \frac{\frac{y}{x} - 2}{\frac{y}{x}} = \frac{y-2x}{y}$$

$$\frac{y-2x}{y} = Cx^2 \quad \text{代} \quad x=1 \text{ 时 } y=1$$

$$-1 = C \Rightarrow C = -1$$

$$\frac{y-2x}{y} = -x^2 \quad \text{解出 } y: \quad y = \frac{2x}{1+x^2}$$

解法二:
$$x^2 y' + xy = y^2$$

$$y' + \frac{1}{x} y = \frac{1}{x^2} (y^2)$$

伯努利方程

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ p(x) & q(x) & \underline{\underline{r(x)}} \end{array}$$

$$z = y^{1-\alpha} = \frac{1}{y}$$

$$z' + (-1) \frac{1}{x} z = (-1) \frac{1}{x^2}$$

$$\underbrace{z' - \frac{1}{x} z}_{p(x)} = \underbrace{-\frac{1}{x^2}}_{q(x)}$$

直接积分法求解。

$$\underline{z = \frac{1}{2x} + Cx = \frac{1}{y}}$$

$$y = \frac{2x}{1+2Cx^2}$$

解 3:

$$x^2 y' + xy = y^2$$

$$\underline{xy' + y} = \frac{1}{x} y^2$$

$$(xy)' = \frac{1}{x} y^2 = \frac{1}{x^3} (xy)^2$$

$$\Delta u = xy.$$

$$\underline{du} = u' = \frac{1}{x^3} u^2$$

$$\frac{1}{u^2} du = \frac{1}{x^3} dx$$

$$\frac{1}{xy} = \frac{1}{u} = \frac{1}{2x^2} + C$$

{ 线性
 伯努利
 可分离
 齐次

例: $\frac{dx}{x^2 - xy + y^2} = \frac{dy}{2y^2 - xy}$ 齐次形式.

$$\frac{dy}{dx} = \frac{2y^2 - xy}{x^2 - xy + y^2} \quad \text{同除 } x^2$$

$$= \frac{2\left(\frac{y}{x}\right)^2 - \frac{y}{x}}{1 - \frac{y}{x} + \left(\frac{y}{x}\right)^2}$$

$$\Delta u = \frac{y}{x}$$

$$\left(x \frac{du}{dx} \right) = \frac{2u^2 - u}{u^2 - u + 1} - u$$

$$= - \frac{u^3 - 3u^2 + 2u}{u^2 - u + 1} = - \frac{u(u-1)(u-2)}{u^2 - u + 1}$$

积分式. $\frac{u^2 - u + 1}{u(u-1)(u-2)} du = - \frac{1}{x} dx$

$$\frac{A}{u} + \frac{B}{u-1} + \frac{C}{u-2} = - \ln x + C$$

$$A = \frac{1}{2} \quad B = -1 \quad C = \frac{3}{2}$$

$$\frac{1}{2} \ln u - \ln(u-1) + \frac{3}{2} \ln(u-2)$$

$$\frac{\sqrt{u} \cdot (u-2)^{\frac{3}{2}}}{u-1} = C \frac{1}{x}$$

$$u = \frac{y}{x}$$

$$(y-x)^2 = Cy (y-2x)^3$$

变量代换:

例: 7.2.1. $y' = \frac{(x+y)(x+y-2)}{x^2}$

$$\text{令 } u = x+y$$

$$\frac{du}{dx} = 1 + \frac{dy}{dx} = \frac{u(u-2)}{x^2} + 1$$
$$= \frac{u^2 - 2u + x^2}{x^2} = \frac{(u-1)^2}{x^2}$$

$$\frac{du}{(u-1)^2} = dx$$

$$-\frac{1}{u-1} = x + C.$$

$$(1-u)(x+C) = 1$$

$$(1-x-y)(x+C) = 1$$

§ 可降阶的高阶微分方程.

注意: $F(x, y, y', \dots, \underbrace{y^{(n)}}) = 0.$

$\underbrace{y^{(n)}} = f(x, y, y', \dots, y^{(n-1)})$

① $y^{(n)} = f(x,$

② $y^{(n)} = f(x, y^{(k)}, \dots, y^{(n-1)}) \quad k \geq 2.$

不显含 $\underbrace{y, y', \dots, y^{(k-1)}}$

③ $y^{(n)} = f(y, y', \dots, y^{(n-1)})$

不显含 x 及 x'

④ $\boxed{y^{(n)}} = f(x)$

不显含 $(y, y', \dots, y^{(n-1)})$

$$dy^{(n-1)} = f(x) dx$$

$$\underline{y^{(n-1)}} = \int f(x) dx + C_1$$

$$dy^{(n-2)} = \left(\underline{\int f(x) dx} + \underline{C_1} \right) dx$$

$$\underline{\underline{y^{(n-2)}}} = \int \left(\int f(x) dx \right) dx + \underline{C_1 x + C_2}$$

$n-1$ 次积分.

⋮

∫ n 次 不定积分

$$y = \underbrace{\int \int \int \int f(x) dx}_{n \text{ 次}} \underbrace{dx \dots dx}_{dx} + \frac{C_1 x^{n-1} + C_2 x^{n-2} + \dots + C_{n-1} x + C_n}{\quad}$$

n 个不定积分

例: 7.3.1. 求 $y'' = 4e^{2x} + \cos x$ 通解.

$$\textcircled{y^{(n)} = f(x)}$$

$$y' = \int 4e^{2x} dx + \int \cos x dx + C_1$$

$$= \underline{2e^{2x}} + \underline{\sin x} + \underline{C_1}$$

$$y = \underline{e^{2x} - \cos x + C_1 x + C_2}$$

$$\underline{\underline{y'' = f(x, y', y)}}_1$$

②. 不含 y 的方程:

$$\underbrace{y'' = f(x, y')}_{(y')'}$$

降阶为 -2π 到 2π .

$$\boxed{z' = f(x, z, t)}$$

$$\underline{\underline{\Delta y' = p(x)}}$$

$$\underline{\underline{p' = f(x, p)}}$$

$$\underline{\underline{p = \varphi(x, C_1)}}$$

$$\frac{dy}{dx} = \varphi(x, C_1)$$

$$y = \int \varphi \cdot + C_2$$

积分常数

1.3.1: $x y'' + y' = 1$ 求通解.

$$y'' = \frac{1}{x} (1 - y')$$

设 $y' = p$

$$\text{令 } y' = p(x)$$

$$y'' + \frac{1}{x} y' = \frac{1}{x}$$

$$p' + \frac{1}{x} p = \frac{1}{x}$$

积分法

$\ln x$

$$p(x) = e^{-\ln x} \left(\int \frac{e^{\ln x}}{x} dx + C_1 \right)$$

$$= x \left(\int \frac{1}{x} dx + C_1 \right)$$

$$= 1 + \frac{C_1}{x}$$

$$\frac{dy}{dx} = 1 + \frac{C_1}{x}$$

$$\underline{y = x + C_1 \ln x + C_2}$$

$$xy'' + y' = 1 \quad y' = p$$

$$\underline{xp' + p = 1}$$

$$(xp)' = 1$$

$$xp = x + C_1$$

$$\underline{p = 1 + \frac{C_1}{x}}$$

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例: 求 $\underline{y'' (x + (y')^2) = y'}$ 初始: $\underline{y(1) = y'(1) = 1}$ 的特解.

令 $y' = p$

$$p' (x + p^2) = p \quad \frac{dp}{dx}$$

$$\Rightarrow$$

$$\frac{dx}{dp} = \frac{x + p^2}{p} = \frac{1}{p} x + p$$

$$\underline{\underline{\frac{dx}{dp} - \frac{1}{p} x = p}}$$

$$\underline{\underline{\frac{1}{p} \frac{dx}{dp} - \frac{1}{p^2} dx = 1}}$$

$$\underline{\underline{\frac{d}{dp} \left(\frac{1}{p} \cdot x \right) = 1}}$$

$$\frac{x}{p} = p + C_1$$

$$x = p^2 + C_1 p$$

$$x=1$$

$$y'=1=p$$

$$1 = 1 + C_1, \quad C_1 = 0. \quad \text{代入 } x=1 \quad p=1$$

$$x = p^2$$

$$p = \sqrt{x}$$

$$y' = \sqrt{x}$$

$$\text{代入 } \begin{cases} x=1 \end{cases}$$

同时结合

$$y = \frac{2}{3} x^{\frac{3}{2}} + C$$

$$C = \frac{1}{3}$$

$$y = \frac{2}{3} x^{\frac{3}{2}} + \frac{1}{3}$$

一般: $y^{(n)} = f(x, \boxed{y^{(k)}, \dots, y^{(n-1)}})$

不显含 $y, y', \dots, y^{(k-1)}$

$$\text{令 } y^{(k)} = u(x)$$

$$y^{(k+1)} = u'$$

$$y^{(k+2)} = u^{(n-k-1)}$$

$$y^{(n)} = u^{(n-k)}$$

n-k阶

$$u^{(n-k)} = f(x, u, \dots, u^{(n-k-1)}),$$

$$\underline{\underline{y^{(k)} = u(x)}}$$

对k求导

例 7.3.2.

$$x y^{(5)} - \underline{y^{(4)}} = 0$$

$$\text{令 } y, y', y'', y'''$$

$$\text{令 } u(x) = y^{(4)}$$

$$x u' = u$$

$$u' - \frac{1}{x} u = 0 \quad \text{线性齐次}$$

$$u = C_1 x$$

$$y^{(4)} = \underline{C_1 x}$$

$$y^{(3)} = \underline{\frac{1}{2} C_1 x^2 + C_2}$$

$$y^{(2)} = \frac{C_1}{6} x^3 + C_2 x + C_3$$

$$y' = \frac{C_1}{24} x^4 + \frac{C_2}{2} x^2 + C_3 x + C_4$$

$$y = \frac{C_1}{120} x^5 + \frac{C_2}{6} x^3 + \frac{C_3}{2} x^2 + C_4 x + C_5$$

$$= \frac{d_1 x^5 + d_2 x^3 + d_3 x^2 + d_4 x + d_5}{59 \text{ 亿 } 2 \text{ 千万}}$$

② 不是含 x 的二次。

$$y'' = f(y, y')$$

$$\text{令 } y' = p(y)$$

→ x 的二次。

$$\frac{d^2 y}{dx^2} = \underline{y''} = \frac{d}{dx}(\underline{p(y)})$$

$$\frac{d}{dx}(y') = \frac{dp}{dy} \cdot \frac{dy}{dx}$$

$$= p \cdot \frac{dp}{dy}$$

$$z' = \varphi(x, z)$$

$$z' = \varphi(y, z)$$

$$\underline{y = y(x)} \quad \underline{y'(x)}$$

$$x = y^{-1}(y) \rightarrow$$

$$p \frac{dp}{dy} = f(y, p) - p \text{ 的二次项}$$

$$\underline{\underline{y' = p(y)}}$$

例 7.3.4. 求 $yy'' - (y')^2 = 0$ (齐次)

$$\text{设 } y' = p(y)$$

$$y'' = p \cdot \frac{dp}{dy}$$

$$y \cdot p \cdot \frac{dp}{dy} - p^2 = 0$$

$$\Leftrightarrow p \left(y \cdot \frac{dp}{dy} - p \right) = 0.$$

↓ 情况 ①: $p=0 \Rightarrow y'=0 \Rightarrow \boxed{y=C}$

↓ 情况 ②: $y \frac{dp}{dy} - p = 0$

$$\Leftrightarrow p' - \frac{1}{y}p = 0 \quad \text{线性齐次方程.}$$

可分离

$$p = C_1 y$$

$$\Rightarrow y' = C_1 y \Rightarrow y = \underline{\underline{C_2 e^{C_1 x}}}$$

例 7.7.5. 上图中一半上凹曲线, $y=y(x)$

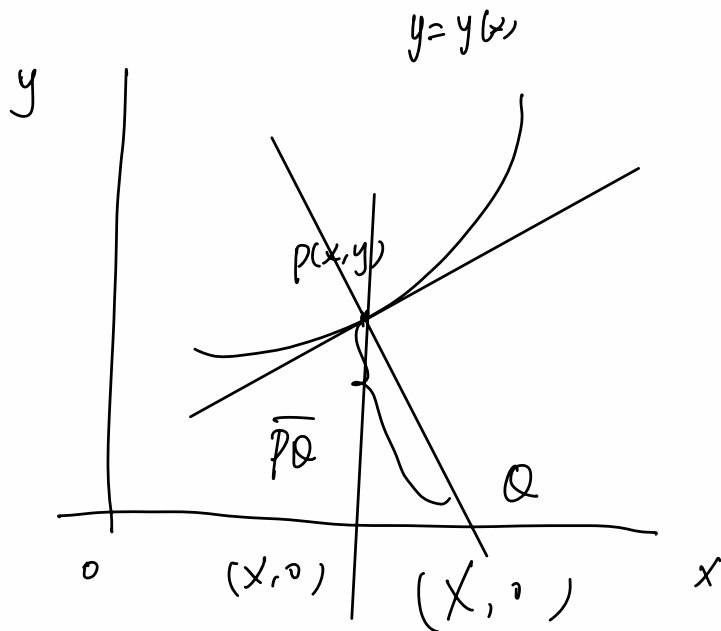
$$y'' > 0.$$

$$k_p = \frac{y''}{[1 + y'^2]^{\frac{3}{2}}}$$

$$k_p = \frac{1}{\rho_Q}$$

已知曲线在点
(1,1) 处切线为
x 轴

求曲线的方程.



已知: $x=1$ 时 $y=1$, $y'(1)=0$ 初始条件.

$$Y-y = y'(X-x) \quad \text{切线}$$

$$Y-y = -\frac{1}{y'}(X-x) \quad \text{法线} \quad (y' \neq 0)$$

$$\text{令 } Y=0 \quad -y = -\frac{1}{y'}(X-x)$$

$$\text{解出 } X = yy' + x \quad Q = (X, 0)$$

$$\begin{aligned} \overline{PQ} &= \sqrt{(x-X)^2 + (y-Y)^2} \\ &= \sqrt{(yy')^2 + y^2} = y \sqrt{1+y'^2} \end{aligned}$$

$$\underline{\underline{\neq y' = 0}}$$

$$\underline{\underline{y''}}$$

$$\underline{\underline{1}}$$

$$[1+y'^2]^{\frac{3}{2}} = y \sqrt{1+y'^2}$$

$$y'' y = 1+y'^2$$

$$\text{Let } y' = p(y) \quad y'' = p \frac{dp}{dy}$$

$$y p \frac{dp}{dy} = 1+p^2$$

$$\frac{p}{1+p^2} dp = \frac{1}{y} dy$$

$$\Rightarrow \frac{1}{2} \ln(1+p^2) = \ln y + C_1$$

$$\sqrt{1+p^2} = C_1 y$$

↓

$$\sqrt{1+p^2} = y \Rightarrow \sqrt{1+y'^2} = y$$

$$y' = \pm \sqrt{y^2-1}$$

$$\frac{dy}{\sqrt{y^2-1}} = \pm dx$$

$$\ln(y + \sqrt{y^2-1}) = \pm (x-1)$$

$$x=1 \text{ at}$$

$$y=1 \text{ at } y'=0$$

$$\Downarrow$$

$$p=0.$$

$$x=1$$

$$y=1$$

$$y = \cosh x = \frac{e^x + e^{-x}}{2}$$

反双曲函数

$$\ln(y + \sqrt{y^2 - 1}) = x$$

$\Downarrow$

$$y = \cosh(\pm(x-1))$$

$$= \frac{1}{2} (e^{x-1} + e^{-(x-1)})$$

④. 不显含 x 的 y 的方程

$$y'' = f(y')$$

例: $y'' + y'^2 = 0$

令 $y' = p(x)$

$$p' + p^2 = 0$$

$$-\frac{dp}{p^2} = dx$$

$$\frac{1}{p} = x + C$$

$$y' = \frac{1}{x+C}$$

$$y = \ln(x+C_1) + C_2$$

或

例: $y'' = (y')^3 + y'$ 的解

$$\textcircled{1} \text{ 不显含 } x \quad y' = p(y) \quad y'' = p \frac{dp}{dy}$$

$$p \frac{dp}{dy} = p^3 + p$$

$$\frac{1}{p^2+1} dp = dy$$

$$\arctan p = y + C_1$$

$$\frac{dy}{dx} = \tan(y + C_1)$$

$$\frac{dy}{\tan(y + C_1)} = dx$$

$$\ln |\sin(y + C_1)| = x + C_2$$

$$\textcircled{2} \text{ 不显含 } y \text{ 的方程} \quad \text{令 } y' = p(x) \quad y'' = p'$$

$$\frac{dp}{dx} = p^3 + p$$

$$\frac{dp}{p(p^2+1)} = dx$$

$$\ln \frac{p}{\sqrt{p^2+1}} = x + C_1$$

$$\frac{p}{\sqrt{p^2+1}} = C_1 e^x$$

$$\text{解出} \quad \frac{dy}{dx} = \pm \sqrt{\frac{C_1^2 e^{2x}}{1 - C_1^2 e^{2x}}}$$

⑤ 小含多导数方程 (连续型)

$$\text{方程} \quad \psi(x, y, \dots, y^{(n)}) = 0.$$

$$\frac{d}{dx} \psi(x, y, \dots, y^{(n-1)}) = 0$$

$$\psi(x, y, \dots, y^{(n-1)}) = C.$$

例: $\underline{yy'' + y'^2 = 0}$

$$\frac{d(yy')}{dy} = 0$$

$$\Rightarrow yy' = C_1$$

$$y dy = C_1 dx$$

$$\frac{1}{2} y^2 = C_1 x + C_2$$

$$y^2 = C_1 x + C_2$$

$$\begin{cases} d(xy) \\ d\left(\frac{x}{y}\right) \\ d(x^2 + y^2) \end{cases}$$

$$\textcircled{(xy)'} =$$

$$d\left(\frac{x}{y}\right) = \frac{y dx - x dy}{y^2}$$

$$\text{例: } \underline{yy'' - y'^2 = 0}$$

解:

$$\frac{yy'' - y'^2}{y^2} = 0$$

$$\frac{d}{dx} \left(\underline{\underline{\frac{y'}{y}}} \right) = 0$$

$$\frac{y'}{y} = C_1$$

$$y = C_2 e^{C_1 x}$$

$$\text{解: } yy'' = y'^2$$

$$\frac{\underline{\underline{(y')'}}}{\underline{\underline{y'}}} = \frac{\underline{\underline{y'}}}{\underline{\underline{y}}}$$

$$\ln y' = \ln y + C_1$$

$$\underline{y' = C_1 y}$$

$$y (y')' - (y)' \cdot (y')$$

$$\frac{y dy' - y' dy}{y^2}$$

§ 线性方程解的结构

非齐次

$$y'' + p(x)y' + q(x)y = f(x)$$

①.

$$f(x) \neq 0$$

齐次

$$y'' + p(x)y' + q(x)y = 0$$

②

n阶

$$y^{(n)} + p_1(x)y^{(n-1)} + \dots + p_n(x)y = f(x)$$

§ 齐次微分方程.

定理 1: 若 $y_1(x)$, $y_2(x)$ 是方程②的两个解, 则 $y(x) = C_1 y_1(x) + C_2 y_2(x)$
也是②的解, (C_1, C_2 任意常数)

$$y_1 = C_1 e^x$$

$$y_2 = 2y_1 = 2C_1 e^x \\ = C_2 e^x$$

$$k_1 y_1 + k_2 y_2 = \underline{\underline{(k_1 + 2k_2) C_1 e^x}}$$

$$= (k_1 + 2k_2) y_1$$

定义: 设 $y_1(x), y_2(x)$ 又在 $I \subseteq \mathbb{R}$. 若 $\exists k \neq 0$, s.t. $y_1 = k y_2$ 恒成 $\forall x \in I$
 称 y_1 和 y_2 为 线性相关; 否则 线性无关.

$\sin x, 2\sin x$ 相关

$\sin x, \cos x$

$$\frac{y_1}{y_2} = C$$

定义: $y_1(x), y_2(x), \dots, y_n(x)$ (1) $n \geq 2$
(2) $(k_1, k_2, \dots, k_n)$
(3) $(y_1, \dots, y_n)$

若有 n 个不全为 0 的常数, $k_1, k_2, \dots, k_n$, s.t.

$$k_1 y_1 + k_2 y_2 + \dots + k_n y_n = 0. \quad (\text{恒为线性相关})$$

$\mathbb{R}, e^x, e^{-x}, e^{2x}$

$1, \cos^2 x, \sin^2 x$

$$1 - \cos^2 x - \sin^2 x = 0$$

$$k_1 = 1 \quad k_2 = -1 \quad k_3 = -1$$

定理: 若 y_1 和 y_2 为方程 (2) 两个线性无关通解, 那么 $y = C_1 y_1 + C_2 y_2$ 就是通解.

例: $y'' + y = 0$

$y_1 = \cos x$ $y_2 = \sin x$

$\frac{y_2}{y_1} = \frac{\sin x}{\cos x} = \tan x \neq \text{常数}$

$y = C_1 \cos x + C_2 \sin x$

线性齐次 n 个特解 线性无关

则: $y_1, y_2, \dots, y_n$

$y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$ 通解.

$y'' + P(x)y' + Q(x)y = 0$

$(x-1)y'' - xy' + y = 0.$

$\left\{ \begin{array}{l} x \\ e^x \end{array} \right.$

$y = C_1 x + C_2 e^x$

① $P(x) + xQ(x) = 0$

$y = x$

故: 线性无关

$$② \quad 1 + p(x) + q(x) = 0$$

$$③ \quad 1 - p(x) + q(x) = 0$$

$$\begin{aligned} y &= e^x \\ y &= e^{-x} \end{aligned}$$

$$y_2 = \textcircled{y_1} \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx$$

§ 非齐次线性方程通解

$$y'' + p(x)y' + q(x)y = f(x) \quad ①$$

$$y'' + p(x)y' + q(x)y = 0 \quad ②$$

性质 1: 设 $y^*(x)$ 是 ① 特解, $\bar{y}$ 是 ② 的通解, 且有 γ 个线性无关,

那么 $y^* + \bar{y} = \underline{\underline{y}}$ 是 ① 的通解.

性质 2. 方程 ① 右端 $f(x) = f_1(x) + f_2(x)$,

$$y_1^* \text{ 是 } y'' + p(x)y' + q(x)y = f_1(x) \text{ 的特解}$$

$$y_2^* \text{ 是 } \quad \quad \quad = f_2(x) \text{ 的特解}$$

$y_1^* + y_2^*$ 是 ① 的特解.

例: 求 $y'' + 4y = x^2 + 3e^x$ 特解.

$$\begin{cases} y'' + 4y = x^2 \Rightarrow y_1^* = \frac{1}{4}x^2 - \frac{1}{8} \\ y'' + 4y = 3e^x \Rightarrow y_2^* = \frac{3}{5}e^x \end{cases}$$

$$y^* = \frac{1}{4}x^2 - \frac{1}{8} + \frac{3}{5}e^x$$

例 7.4.2 若 y_1, y_2 是 ① 的两个特解, $y_1 - y_2$ 是 ② 的特解.

例 7.4.1.

$$\begin{cases} y_1 = xe^x + e^{2x} \\ y_2 = xe^x + e^{-x} \\ y_3 = \boxed{xe^x} + e^{2x} - e^{-x} \end{cases}$$

$$y = \underbrace{y_x}_{\substack{\downarrow \\ \text{非齐次} \\ \text{特解}}} + C_1 \underbrace{y_1}_{\substack{\downarrow \\ \text{齐次} \\ \text{特解}}} + C_2 \underbrace{y_2}_{\substack{\downarrow \\ \text{齐次} \\ \text{特解}}}$$

$$\underline{y^*} = \underline{xe^x}$$

$$\boxed{y_1 - y_3} = e^{-x}$$

$$\boxed{y_1 - y_2} = e^{2x} - e^{-x}$$

$$\boxed{e^{2x}}$$

$\frac{2}{5}$ 是 ② 的特解.

$$y(x) = xe^x + C_1 e^{-x} + C_2 e^{2x}$$

$$y'(x) = e^x + xe^x - C_1 e^{-x} + 2C_2 e^{2x}$$

$$y''(x) = 2e^x + xe^x + C_1 e^{-x} + 4C_2 e^{2x}$$

$$y'' - y' - y = e^x - 2xe^x.$$

§7.5 常系数齐次线性方程

$$y'' + py' + qy = 0$$

$$\underline{y(x) = e^{rx}} \quad (r \text{ 为常数}) \quad y' = re^{rx} \quad y'' = r^2 e^{rx}$$

$$\underline{(r^2 + pr + q)e^{rx} = 0, \quad e^{rx} \neq 0}$$

$$\Rightarrow \begin{cases} r^2 + pr + q = 0. \end{cases} \quad \left(\begin{array}{l} \text{二次} \\ \text{代数方程} \end{array} \right)$$

4种情况.

$$y'' + py' + qy = 0 \quad \checkmark \quad \text{特征根.}$$

$$r_{1,2} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

$$\textcircled{1} \quad p^2 - 4q > 0. \quad r_1 \neq r_2$$

$$\begin{cases} y_1 = e^{r_1 x} \\ y_2 = e^{r_2 x} \end{cases} \quad \frac{y_1}{y_2} = e^{(r_1 - r_2)x} \neq 0.$$

线性无关

$$\Rightarrow \text{通解 } y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

$$\text{验证: 设 } y_1 = e^{r_1 x} \text{ 和 } y_2 = e^{r_2 x}$$

$$r_1 \neq r_2 \text{ 且 } \overline{r_1} \neq r_2.$$

$$(r - r_1)(r - r_2) = 0$$

$$r^2 - (r_1 + r_2)r + r_1 r_2 = 0.$$

$$y'' - (r_1 + r_2)y' + r_1 r_2 y = 0$$

$$\textcircled{2} p^2 - 4q = 0.$$

二重根

$$r = -\frac{p}{2}$$

$$(x + \frac{p}{2})^2 = 0$$

$$\text{一个特解: } y_1 = e^{rx}$$

常数变易法

$$C y_1(x)$$

$$C \rightarrow h(x)$$

$$y_2 = h(x, y_1(x))$$

$$y_2' = h'(x) e^{rx} + r h(x) e^{rx}$$

$$y_2'' = h''(x) e^{rx} + r h'(x) e^{rx} + r h'(x) e^{rx} + r^2 h(x) e^{rx}$$

$$y_2'' + p y_2' + q y_2 = 0$$

$$\begin{aligned} & \underline{h'' e^{rx}} + 2r \underline{h' e^{rx}} + \underbrace{r^2 h e^{rx} + p r h e^{rx} + q h e^{rx}}_{=0} \\ & + p \underline{h' e^{rx}} \end{aligned}$$

$$r^2 + pr + q = 0$$

$$e^{rx} \neq 0$$

$$\underline{h'' + (2r+p)h' = 0}$$

(+ 特解)

$$(\text{韦达定理}) \quad r = -\frac{p}{2} \Rightarrow 2r + p = 0.$$

$$h'' = 0 \Rightarrow \text{取 } h(x) = x$$

$$\text{第二个特解} \quad \begin{cases} y_2 = x e^{rx} \\ y_1 = e^{rx} \end{cases}$$

$$y(x) = C_1 y_1 + C_2 y_2 = e^{rx} (C_1 + C_2 x)$$

$$\text{已知 } \underline{y = x e^{rx}} \quad \text{是一个特解.}$$

$$(r - r_1)^2 = 0$$

$$r^2 - 2r_1 r + r_1^2 = 0$$

$$\downarrow \quad y'' - 2r_1 y' + r_1^2 y = 0$$

$$\textcircled{3} \quad p^2 - 4q < 0 \quad \text{有两个共轭复根.}$$

$$r_1 = \alpha + i\beta \quad r_2 = \alpha - i\beta$$

$$\begin{cases} y_1 = e^{r_1 x} = e^{(\alpha + i\beta)x} = e^{\alpha x} \cdot e^{i\beta x} \\ y_2 = e^{r_2 x} = e^{(\alpha - i\beta)x} = e^{\alpha x} \cdot e^{-i\beta x} \end{cases}$$

$$y(x) = e^{\alpha x} (D_1 e^{i\beta x} + D_2 e^{-i\beta x})$$

($x \in \mathbb{R}$)

$$e^{ix} = \cos x + i \sin x \quad \text{欧拉公式.}$$

$$y_1 = e^{\alpha x} (\cos \beta x + i \sin \beta x)$$

$$y_2 = e^{\alpha x} (\cos \beta x - i \sin \beta x)$$

实部和虚部提取出来.

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2i} & -\frac{1}{2i} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(y_1 + y_2) \\ \frac{1}{2i}(y_1 - y_2) \end{pmatrix}$$

$$\bar{y}_1 = \frac{1}{2}(y_1 + y_2) = e^{\alpha x} \cos \beta x$$

$$\bar{y}_2 = \frac{1}{2i} (y_1 - y_2) = e^{\alpha x} \sin \beta x$$

$$y(x) = C_1 \bar{y}_1 + C_2 \bar{y}_2 = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

① $y'' + py' + q = 0$
 $\downarrow$ 特征方程

$$r^2 + pr + q = 0$$

② 求根 $r_1 \quad r_2$

③ 有二实根 $r_1 \neq r_2$ $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$

有二重根 r $y = (C_1 + C_2 x) e^{rx}$

有共轭复根

$$r_{1,2} = \alpha \pm i\beta$$

$$y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

例: 求 $y'' - 2y' - 3y = 0$ 通解

$$r^2 - 2r - 3 = 0$$

$$(r-3)(r+1) = 0 \quad r_1 = -1 \quad r_2 = 3$$

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x} = C_1 e^{-x} + C_2 e^{3x}$$

例: 求 $y'' + 4y' + 4y = 0$ 通解.

$$r^2 + 4r + 4 = 0$$

$$(r+2)^2 = 0 \quad \text{二重根} \quad r = -2$$

$$y = (C_1 + C_2 x) e^{rx} = (C_1 + C_2 x) e^{-2x}$$

例: 求 $y'' + 2y' + 5y = 0$ 通解.

$$r^2 + 2r + 5 = 0 \quad \sqrt{p^2 - 4q} = \sqrt{-4}$$

$$r_1 = -1 + 2i \quad \left\{ \begin{array}{l} \alpha = -1 \end{array} \right.$$

$$r_2 = -1 - 2i \quad | \quad \beta = 2$$

$$y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

$$= e^{-x} (C_1 \cos 2x + C_2 \sin 2x)$$

例 7.5.4. 解 $yy'' - y'^2 = y^2 \ln y$

(不是全x的分程) $\Delta y' = p(y)$

$$y'' = p \frac{dp}{dy}$$

凑微分

$$\frac{yy'' - y'^2}{y^2} = \ln y$$

$$\left(\frac{y'}{y}\right)' = \ln y$$

$$((\ln y)')'$$

$$(\ln y)'' = \ln y$$

$$\Delta z = \ln y$$

$$z' - z = 0$$

$$r^2 - 1 = 0 \quad r = \pm 1$$

$$z = \ln y = \underline{C_1 e^x + C_2 e^{-x}}$$

例: 已知 $x = \sinh t$ 是特解. 求微分方程.

$$x = \sinh t = e^{0 \cdot x} \sinh 1 \cdot x$$

$$\alpha = 0$$

$$\beta = 1$$

$$\Rightarrow r_{1,2} = \pm i$$

$$(r-i)(r+i) = 0$$

$$r^2 + 1 = 0$$

↓

$$y'' + y = 0$$

n 阶常系数齐次微分方程

$$y^{(n)} + p_{n-1} y^{(n-1)} + p_{n-2} y^{(n-2)} + \dots + p_1 y' + p_0 y = 0.$$

↓ 特征方程

$$P_n = r^n + p_1 r^{n-1} + p_2 r^{n-2} + \dots + p_{n-1} r + p_n = 0$$

(n 次代数方程) n 个根.

二重根: 二重根

$$P_n = (x-r)^k \underline{\underline{R_{n-k}'}} \quad \underline{\underline{k=1}} \quad \underline{\underline{k \geq 2}}$$

$$\boxed{C_1 e^{rx}}$$

$$\underline{e^{rx} (C_1 + C_2 x)}$$

$$\underline{e^{rx} (C_1 + C_2 x + \dots + C_k x^{k-1})}$$

k 个独立
第 k 个

$$P_n = \boxed{(x-a)^{k_1}} \dots (x-b)^{k_2} \underline{(x^2+px+q)^k} \dots (x^2+cx+d)^{r_2}$$

$$k_1 + \dots + k_2 + 2(r_1 + \dots + r_2) = n$$

$$\underline{c^2 - 4d < 0}$$

实根

一对共轭复根

2 个

$$\boxed{\text{复根}}$$

$$V_{1,2} = \alpha \pm i\beta$$

$$e^{\alpha x} \left(\underline{C_1 \cos \beta x} + \underline{C_2 \sin \beta x} \right)$$

k 个复根

2k 个

k-1 个复根

$$e^{\alpha x} \left((C_1 + C_2 x + \dots + C_k x^{k-1}) \cos \beta x + (D_1 + D_2 x + \dots + D_k x^{k-1}) \sin \beta x \right)$$

例: 解 $y^{(4)} - 2y'' + 5y = 0$ 解.

$$r^4 - 2r^2 + 5 = 0.$$

$$\Rightarrow r^2(r^2 - 2r + 5) = 0.$$

二重实根 $r=0$

共轭复根 $r_3 = 1+2i$
 $r_4 = 1-2i$

$$e^{rx} (C_1 + C_2 x) = C_1 + C_2 x$$

$\alpha=1 \quad \beta=2$

$$e^{\alpha x} (C_3 \cos \beta x + C_4 \sin \beta x)$$

$$e^x (C_3 \cos 2x + C_4 \sin 2x)$$

$$y(x) = C_1 + C_2 x + e^x (C_3 \cos 2x + C_4 \sin 2x)$$

§ 常系数非齐次

$$y'' + py' + qy = f(x) \quad (1)$$

(p, q 为常数, $f(x) \neq 0$)

$$y = y^* + \bar{y}$$

$\downarrow$
齐次 ... 通解

待定系数

$$r^2 + pr + q = 0. \quad r_{1,2}$$

$$\begin{cases} r_1 + r_2 = -p \\ r_1 r_2 = q \end{cases}$$

$$y'' - \underline{(r_1 + r_2)} y' + r_1 r_2 y = f(x),$$

$$\Rightarrow \underline{\underline{y'' - r_1 y'}} - \underline{\underline{r_2 y' + r_1 r_2 y}} = f(x)$$
$$\underline{\underline{r_2 (y' - r_1 y)}} = f(x)$$

$$\Rightarrow (y' - r_1 y)' - r_2 (y' - r_1 y) = f(x) \Rightarrow u' - r_2 u = f(x),$$

$$\underline{\Delta \quad u(x) = y' - r_1 y}$$

$$\begin{cases} u' - r_2 u = f(x) & (2) \\ y' - r_1 y = u(x) & (3) \end{cases}$$

$$(2) \Rightarrow u(x) = C_2 e^{r_2 x} + e^{r_2 x} \int e^{-r_2 x} f(x) dx$$

$$(3) \Rightarrow y(x) = C_1 e^{r_1 x} + C_2 e^{r_1 x} \int e^{(r_2 - r_1)x} dx \\ + e^{r_1 x} \int e^{(r_2 - r_1)x} \left(\int e^{-r_2 x} f(x) dx \right) dx$$

例1: 解 $y'' - 4y' + 4y = \frac{e^{2x}}{x}$

$$r^2 - 4r + 4 = 0 \quad r_1 = r_2 = 2.$$

$$u = C_1 e^{2x} + C_2 e^{2x} \int e^{0x} dx$$

$$+ e^{2x} \int e^{0x} \left(\int \underline{e^{-2x} \frac{e^{2x}}{x} dx} \right) dx$$

$$= C_1 e^{2x} + C_2 x e^{2x} + e^{2x} \int \underline{1 \cdot \ln|x| dx}$$

$$= C_1 e^{2x} + C_2 x e^{2x} + e^{2x} (x \ln|x| - x)$$

特解 $f(x)$,

$$\textcircled{1} \quad f(x) = \underline{e^{\lambda x} P_m(x)}$$

$$\textcircled{2} \quad f(x) = \underline{e^{\lambda x} [P_l(x) \cos wx + P_n(x) \sin wx]}$$

第 一 类:

$$y'' + py' + qy = e^{\lambda x} P_m(x)$$

$$(特解) \quad e^{\lambda x} Q_n(x) = e^{\lambda x} (b_0 + b_1 x + \dots + b_n x^n)$$

$$\text{设 } y(x) = e^{\lambda x} Q_n(x) \text{ 且 } \lambda \text{ 不是根:}$$

$$y' = \lambda e^{\lambda x} Q_n + e^{\lambda x} Q_n'$$

$$y'' = \lambda^2 e^{\lambda x} Q_n + 2\lambda e^{\lambda x} Q_n' + e^{\lambda x} Q_n''$$

$$\Rightarrow \underbrace{Q_n''}_{n-2} + (2\lambda + p) \underbrace{Q_n'}_{n-1} + (\lambda^2 + p\lambda + q) \underbrace{Q_n}_{n} = \underbrace{P_m}_{m}$$

$$\textcircled{1} \quad \text{若 } \lambda \text{ 不是特征方程的根.} \quad \lambda^2 + p\lambda + q \neq 0.$$

$$\text{二) } n = m$$

$$\Rightarrow y^* = e^{\lambda x} Q_m(x)$$

$$\textcircled{2} \quad \lambda \text{ 是单根.} \quad \lambda^2 + 2\lambda + p = 0$$

$$2\lambda + p \neq 0$$

$$n = m+1 \quad \text{if } Q_n(x) = x Q_m(x)$$

$$y^* = x e^{\lambda x} Q_m(x)$$

② λ is a root.

$$\lambda^2 + p\lambda + q = 0$$

$$2\lambda + p = 0$$

$$\Rightarrow Q_n'' = P_m(x), \quad n = m+2$$

$$Q_n(x) = x^2 P_m(x)$$

$$y^* = x^2 Q_m(x) e^{\lambda x}$$

① $f(x) = e^{\lambda x} P_m(x)$

λ is not a root

λ is a root

λ is a double root

$$y^* = e^{\lambda x} Q_m(x)$$

$$y^* = x e^{\lambda x} Q_m(x)$$

$$y^* = x^2 e^{\lambda x} Q_m(x)$$

}

$$\textcircled{2} f(x) = e^{\lambda x} \left[\underbrace{P_l(x)}_{\lambda \pm i\omega} \cos \omega x + \underbrace{P_n(x)}_{m = \max\{l, n\}} \sin \omega x \right].$$

$$\textcircled{1} \text{ 不是特征根 } y^* = e^{\lambda x} [Q_m(x) \cos \omega x + R_m(x) \sin \omega x]$$

$$\textcircled{2} \text{ 是特征根 } y^* = x e^{\lambda x} [\quad \quad \quad].$$

例: 解 $y'' + y' = x^2$

$$r^2 + r = 0 \quad r_1 = 0 \quad r_2 = -1$$

(齐次) $\bar{y} = C_1 + C_2 e^{-x}$

求特解: $x^2 = e^{0 \cdot x} x^2 = e^{0 \cdot x} \cdot P_2(x)$ 形式.

$\lambda = 0$ 是一个特征根.

$$\text{if } y^* = x e^{\lambda x} Q_2(x) \\ = x (ax^2 + bx + c)$$

$$(y^*)' = 3ax^2 + 2bx + c$$

$$(y^*)'' = 6ax + 2b$$

$$3ax^2 + (2b + 6a)x + (c + 2b) = x^2$$

$$\begin{cases} a = \frac{1}{3} \\ b = -1 \\ c = 2 \end{cases}$$

$$y = \bar{y} + y^* = C_1 + C_2 e^{-x} + \frac{1}{3}x^3 - x^2 + 2x$$

12.1: $\ddot{y} - 4\dot{y} + 4y = \underbrace{6x^2 + 8e^{2x}}_{\text{RHS}}$

$$\begin{cases} y'' - 4y' + 4y = 6x^2 & Y_1 \\ - - - = 8e^{2x} & Y_2 \end{cases}$$

$$Y^* = Y_1 + Y_2$$

$$r^2 - 4r + 4 = 0 \quad r = 2 \quad \text{二重根}$$

$$\textcircled{1} f(x) = 6x^2 = e^{0 \cdot x} \cdot P_2(x) \quad \lambda = 0 \quad \text{不是特征根.}$$

$$Y_1 = e^{0 \cdot x} Q_2(x) = Ax^2 + Bx + C$$

$$\begin{cases} A = \frac{3}{2} \\ B = 3 \\ C = 15 \end{cases}$$

$$Y_1 = \frac{3}{2}x^2 + 3x + 15$$

$$\textcircled{2} f(x) = 8e^{2x} = e^{2 \cdot x} \cdot P_0(x)$$

$$\lambda = 2 \quad \text{二重根.}$$

$$\text{设特解 } Y_2 = x^2 e^{2 \cdot x} \cdot Q_0(x),$$

$$= Dx^2 e^{2x}$$

$$Y_2' = 2Dx e^{2x} + 2Dx^2 e^{2x}$$

$$\underline{Y_2''} = 2D e^{2x} + 8Dx e^{2x} + 4Dx^2 e^{2x}$$

$$2D e^{2x} = 8 e^{2x}$$

$$D = 4$$

$$Y_2 = 4x^2 e^{2x}$$

$$Y^* = Y_1 + Y_2 = \frac{3}{2}x^2 + 3x + 15 + 4x^2 e^{2x}$$

$$f(x) = 4 \sinh x$$

$$\underline{y'' + y = 4 \sinh x}$$

$$r^2 + 1 \quad r = \pm i \quad \begin{cases} \alpha = 0 \\ \beta = 1 \end{cases} \quad 0 \pm i$$

$$4 \sinh x = e^{\underline{0 \cdot x}} \left[\underline{0} \cdot \cos x + \underline{4} \cdot \sin x \right]$$

$$y^* = x \boxed{e^{0 \cdot x}} [A \cos x + B \sin x].$$

$$= x [A \cos x + B \sin x].$$

§ 常变易法.

一阶: $\frac{dy}{dx} + p(x)y = q(x) \quad q(x) \neq 0.$

$$y = \bar{y} + y^*$$

$$\frac{dy}{dx} + p(x)y = 0$$

$$\frac{dy}{y} = -p(x) dx$$

$$\ln y = -\int p(x) dx + C$$

$$y = C e^{-\int p(x) dx}$$

$$\triangle C \rightarrow u(x)$$

$$y = u(x) e^{-\int p(x) dx}$$

↓ 不是常数.

$$\frac{dy}{dx} = u'(x) e^{-\int p(x) dx} + u(x) \left(-e^{-\int p(x) dx} \cdot p(x) \right)$$

$$u'(x) e^{-\int p(x) dx} - \underbrace{u p e^{-\int p(x) dx}} + \underbrace{u p e^{-\int p(x) dx}} = q(x),$$

$$\frac{dy}{dx} e^{-\int p(x) dx} = q(x) \quad \text{两边} \int$$

$$du = e^{\int p(x) dx} q(x) dx$$

$$u(x) = \int e^{\int p(x) dx} q(x) dx$$

$$y = e^{-\int p(x) dx} \left(\int e^{\int p(x) dx} q(x) dx + C \right)$$

二阶 齐次线性方程.

$$y'' + p(x)y' + q(x)y = 0 \quad \text{D.}$$

$$y_1(x) \quad \text{特解.} \quad C_1 y_1(x) \quad C_2 \text{ 常数.}$$

$$\hat{A} C \rightarrow u(x),$$

$$y(x) = u(x) y_1(x) \text{ 也 } \frac{1}{y_1} \text{ 解.}$$

$$y'(x) = u'(x) y_1(x) + u(x) y_1'(x),$$

$$y''(x) = u''(x) y_1(x) + 2u'(x) y_1'(x) + u(x) y_1''(x),$$

$$u'' y_1 + \underline{2u' y_1'} + \underline{u y_1''} + \underline{p(u' y_1 + u y_1')} + \underline{q u y_1} = 0.$$

$$y_1 u'' + (2 y_1' + p y_1) u' = 0. \quad \bar{q} \text{ 舍去}$$

$$\frac{u''}{u'} = - \frac{2 y_1' + p y_1}{y_1} = - \frac{2 y_1'}{y_1} - p$$

$$\ln u' = -2 \ln y_1 - \int p(x) dx$$

$$\ln \frac{1}{y_1^2}$$

$$u' = \frac{1}{y_1^2} e^{-\int p(x) dx}$$

$$u(x) = \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx$$

∴ 1/3 式.

$$y_2 = u y_1 = y_1(x) \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx \quad \begin{cases} \text{如果有 } C \\ \text{没有 } C, \end{cases}$$

$$y = \underline{C_1} y_1 + \underline{C_2} y_2$$

例 7.7.1.

$$y'' - 4xy' + (4x^2 - 2)y = 0.$$

已知一个特解 $y_1 = e^{x^2}$, 求通解.

解:

$$p(x) = -4x$$

$$q(x) = 4x^2 - 2$$

$$y_2(x) = e^{x^2} \int \frac{1}{(e^{x^2})^2} e^{-\int (-4x) dx} dx$$

$$= e^{x^2} \int \frac{e^{-2x^2}}{e^{2x^2}} e^{2x^2} dx$$

$$= x e^{x^2}$$

$$y = C_1 y_1 + C_2 y_2 = C_1 e^{x^2} + C_2 x e^{x^2}$$

$$= e^{x^2} (C_1 + C_2 x)$$

验证
二阶非齐次

$$y'' + p(x)y' + q(x)y = f(x), \quad (3) \quad f(x) \neq 0$$

已知齐次方程 $(y'' + p(x)y' + q(x)y = 0)$

$$\text{通解: } Y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$\wedge \quad C_1 \rightarrow u_1(x)$$

$$C_2 \rightarrow u_2(x)$$

$$Y(x) = u_1 y_1 + u_2 y_2$$

$$Y' = \underline{u_1' y_1 + u_1 y_1'} + u_2' y_2 + u_2 y_2'$$

$$Y'' = \underline{u_1'' y_1 + 2 u_1' y_1' + u_1 y_1''}$$

$$+ u_2'' y_2 + 2 u_2' y_2' + u_2 y_2''$$

$$u_1'' y_1 + 2 u_1' y_1' + u_1 y_1'' + p(u_1' y_1 + u_1 y_1') + q(u_1 y_1)$$

$$+ \dots = f(x)$$

$$\Rightarrow u_1'' y_1 + 2 u_1' y_1' + p u_1' y_1 + u_2'' y_2 + 2 u_2' y_2' + p u_2' y_2 = f(x)$$

$$\underbrace{(u_1'' y_1 + u_2'' y_2)}_{\text{two terms cancel}} + \underbrace{2(u_1' y_1' + u_2' y_2')}_{u_1' y_1 + u_2' y_2 = 0} + p(u_1' y_1 + u_2' y_2) = f(x)$$

$$\underline{y_1 u_1'' + y_1' u_1' + y_1' u_1'}$$

$$(y_1 u_1')'$$

$$\underline{(y_1 u_1' + y_2 u_2')}' + (y_1' u_1' + y_2' u_2') = f(x)$$

0

$$y_1 u_1' + y_2 u_2' = 0$$

$$\begin{cases} y_1' u_1' + y_2' u_2' = f(x), \end{cases}$$

$$\underline{\underline{\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ f(x) \end{pmatrix}}}$$

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0.$$

$$[\text{假设. } \underline{y_1 y_2' - y_2 y_1' = 0}]$$

$$\frac{y_2'}{y_2} = \frac{y_1'}{y_1}$$

$$\ln y_2 = \ln y_1 + C$$

$$\underline{y_2 = C y_1}$$

 (假设和反证法) 矛盾

$$\begin{cases} u_1'(x) = - \frac{y_2(x) f(x)}{W(x)} \\ u_2'(x) = \frac{y_1(x) f(x)}{W(x)} \end{cases}$$

$$\int u_1(x) = - \int \dots dx$$

$$y^* = u_1 y_1 + u_2 y_2$$

$$u(x) = \int \dots dx$$

$$y = \bar{y} + y^*$$

$$= C_1 y_1 + C_2 y_2 + u_1 y_1 + u_2 y_2$$

其中

任意常数

$$(C_1) y_1 + \dots + (C_n) y_n$$

$$u_1(x) y_1 + \dots + u_n(x) y_n$$

$$\begin{pmatrix} \cdot & \cdot & \cdot \\ & \ddots & \\ & & \cdot \end{pmatrix}_{n \times n} \begin{pmatrix} u_1' \\ \vdots \\ u_n' \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ f \end{pmatrix}$$

例: 7.7.2.

$$y'' + \frac{x}{1-x} y' - \frac{1}{1-x} y = x-1$$

求

$$y'' + \frac{x}{1-x} y' - \frac{1}{1-x} y = 0 \quad \text{通解}$$

↑

$$y_1(x) = e^x$$

$$y_2 = e^x \int \frac{1}{(e^x)^2} e^{-\int \frac{x}{1-x} dx} dx$$

$$= e^x \int \underbrace{e^{-2x} e^x}_{e^{-x}} (x-1) dx$$

$$= e^x \int e^{-x} (x-1) dx$$

$$= \underline{\underline{-x}}$$

$$y_2 = x \text{ 也是解}$$

$$\bar{y} = c_1 x + c_2 e^x$$

$$y_1 = e^x$$

$$y_2 = x$$

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x \\ e^x & 1 \end{vmatrix} = e^x - xe^x \neq 0$$

$$y^* = -e^x \int \frac{x(x-1)}{e^x(x-1)} dx + x \int \frac{e^x(x-1)}{e^x(x-1)} dx$$

$$= -x^2 - x - 1$$

$$y = \bar{y} + y^*$$

$$= \textcircled{C_1 x} + C_2 e^x - x^2 - 1$$

§ 2.8. 欧拉-柯西方程.

定义: 一般形式:

$$x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_1 x y' + a_0 y = f(x),$$

$f(x)=0$. 欧拉-柯西齐次方程.

§. 求解 欧拉-柯西方程 (齐次)

$$\textcircled{x^2 y'' + b x y' + c y = 0} \quad r \in \mathbb{C}.$$

设解: $y(x) = x^r \quad y' = r x^{r-1} \quad y'' = r(r-1) x^{r-2}$

$$r(r-1) x^r + b r x^r + c x^r = 0.$$

$$r(r-1) + b r + c = 0.$$

$$r^2 + (b-1)r + c = 0.$$

$$r_{1,2} = \frac{-(b-1) \pm \sqrt{(b-1)^2 - 4c}}{2}$$

① $r_1 \neq r_2$ 两个不同的根.

$$y(x) = C_1 x^{r_1} + C_2 x^{r_2}$$

② $r = \frac{1}{2}$ 重根. $y_1 = C x^r$ $C \rightarrow u(x)$

$$\underline{y_2 = u(x) x^r}$$

$$\underline{y_2'} = u' x^r + r u x^{r-1} = (u' x + r u) x^{r-1}$$

$$\underline{y_2''} = u'' x^r + 2 r u' x^{r-1} + r(r-1) u x^{r-2}$$

$$= x^{r-2} (u'' x^2 + 2 r x u' + r(r-1) u)$$

$$x^2 u'' + 2 r x u' + \underline{r(r-1) u}$$

$$+ b x u' + \underline{b r u}$$

$$+ \underline{c u} = 0.$$

$$\rightarrow x^2 u'' + (2 r x + b) u' + (r(r-1) + b r + c) u = 0$$

$$2 r = 1 - b$$

$$\Rightarrow x^2 u'' + 2x u' + 5u = 0$$

$$2r + b = 1$$

$$\Rightarrow x^2 u'' + x u' = 0$$

$$\frac{u''}{u'} = -\frac{1}{x}$$

$$\ln|u'| = -\ln(x)$$

$$u' = \frac{1}{x}$$

$$u = \ln x$$

$$y = C_1 x^r + C_2 x^r \ln x$$

$$\textcircled{3}. \quad V_{1,2} = \alpha \pm i\beta \quad x^{\alpha \pm i\beta}$$

$$x^{\alpha + i\beta} = e^{(\alpha + i\beta) \ln x}$$

$$= \underbrace{e^{\alpha \ln x}} e^{i\beta \ln x}$$

$$= x^\alpha (\cos(\beta \ln x) + i \sinh(\beta \ln x))$$

$$y(x) = x^\alpha (C_1 \cos(\beta \ln x) + C_2 \sinh(\beta \ln x))$$

法二. 变量代换法.

作变量代换: $x = e^t$ 则 $t = \ln x$.

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \Rightarrow x \frac{dy}{dx} = \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d \left(\frac{dy}{dt} \right)}{dx}$$

$$= -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d \left(\frac{dy}{dt} \right)}{dt} \cdot \frac{dt}{dx}$$

$$= -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x^2} \frac{d^2y}{dt^2}$$

$$\Rightarrow x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \right)$$

$$= \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right)$$

$$\Rightarrow \quad \underline{x^3 \frac{d^3 y}{dx^3}} = \frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt}$$

$$\bigcirc \quad \boxed{x^k \frac{d^k y}{dx^k}} = \frac{d^k y}{dt^k} + \underbrace{\bigcirc a_k}_{\leftarrow z_k} \frac{d^{k-1} y}{dt^{k-1}} + \dots + \bigcirc a_1 \frac{dy}{dt}$$

$$\frac{y}{x} \propto \cos.$$

$$x = -e^t$$

$$t = \ln(-x)$$

$$12.1: \quad t^2 y'' + 3t y' + y = 0$$

$$\begin{cases} y(1) = 0 \\ y'(1) = 1 \end{cases}$$

12.4:

$$b=3 \quad c=1$$

$$y = t^r$$

$$r(r-1) + 3r + 1 = 0$$

$$\underline{r^2 + 2r + 1 = 0}$$

$$r = -1 \quad \text{二重根}$$

$$y = (C_1 + C_2 \ln |t|) \frac{1}{t}$$

$$t=1 \quad y=0 \quad \Rightarrow \quad C_1 = 0$$

$$y = C_2 \ln |t| \frac{1}{t} \quad y' = C_2 \frac{1 - \ln |t|}{t^2}$$

$$t=1 \quad y'=1 \quad C_2 = 1$$

$$y(t) = \frac{\ln |t|}{t}$$

注::

$$t^2 y'' = \frac{d^2 y}{dx^2} - \frac{dy}{dx}$$

$$t y' = \frac{dy}{dx}$$

$$\frac{d^2 y}{dx^2} - \frac{dy}{dx} + 3 \frac{dy}{dx} + y = 0.$$

$$y'' + 2y' + y = 0.$$

例. $x^2 y'' - 2xy' + 2y = \ln^2 x - 2\ln x$ 通解.

齐次: $x = e^t \quad t = \ln x$

$$\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2\frac{dy}{dt} + 2y = t^2 - 2t$$

$$(y'' - 3y' + 2y = t^2 - 2t) \quad (\text{齐次})$$

$$r^2 - 3r + 2 = 0$$

$$r=1 \quad r=2.$$

齐次: $y(t) = C_1 e^t + C_2 e^{2t}$

$$f(t) = \underline{e^{0 \cdot t}} \quad (P_2(t)) \quad \lambda=0 \text{ 不是特征根.}$$

$$\text{设: } y^* = \underline{x^0 \cdot e^{0 \cdot t} Q_2(t)}$$

$$at^2 + bt + c \quad \text{代入.}$$

$$(y^*)' = 2at + b$$

$$(y^*)' = 2at$$

$$(y^*)'' = 2a$$

$$2a - 3(2at + b) + 2(at^2 + bt + c) = t^2 - 2t$$

$$\begin{cases} a = \frac{1}{2} \\ b = \frac{1}{2} \\ c = \frac{1}{4} \end{cases}$$

$$y = C_1 e^t + C_2 e^{2t} + \frac{1}{2} t^2 + \frac{1}{2} t + \frac{1}{4}$$

$$\text{代入 } t = \ln x$$

$$y = C_1 x + C_2 x^2 + \frac{1}{2} \ln^2 x + \frac{1}{2} \ln x + \frac{1}{4}$$

§5.6, 5.7, §6.4, §7.8, 7.9

$\left\{ \begin{array}{l} \text{极限} \\ \text{导数/微分} \\ \text{积分} \end{array} \right.$

$\left\{ \begin{array}{l} \text{定义} \\ \text{性质 (定理/证明)} \\ \text{应用} \end{array} \right.$

100分 笔试 120分钟

90% 计算题 10% 证明题

五道大题

一. 求极限

二. 求导数

三. 求级数

四. 应用

五. 证明题

一. 求极限. $\left\{ \begin{array}{l} \text{七种极限过程} \\ \text{无穷小/无穷大} \end{array} \right.$ 唯一性 (局部性)

运算法则.

微分法则 $\left\{ \begin{array}{l} \text{三阶法公式} \\ \text{高阶有界原理} \end{array} \right.$

Heine 定理

$(1 + \frac{1}{n})^n$

数列极限.

小-ε 定义

无穷小-ε 定义

极限

Taylor 展开

合数定义. / (x) / $\sqrt[n]{x}$

§1.7 §1.8

求导. 幂指函数 $[f(x)]^{g(x)}$

二. 导数.

微分法则

四则运算

复合函数

反函数

高阶导数

隐函数求导

(参数方程求导)

§2-5:

若被积函数为有理函数 = 部分分式法

例 2.5.3

待定系数法. $y = \frac{u(x), v(x), \sqrt[n]{x}}{x+1, 3}$

$$\ln|y| = \quad - \quad - \quad -$$

$$\frac{d \ln|x|}{dx} = \frac{1}{x} \quad \text{待定}.$$

积分: 不定积分 (C)

积分法则 $\left\{ \begin{array}{l} \text{合积} \\ \text{合积} \\ -1 = \text{换元法} \end{array} \right.$

$$\begin{array}{lll} \int \tan x dx & \int \sec x dx & \int \sec^2 x dx \\ & \int \sin^2 x dx & \int \cos^2 x dx \end{array}$$

$$\left\{ \int \frac{1}{a^2 - x^2} dx \quad \int \sqrt{a^2 - x^2} dx \right. \\ \left. (\text{三角代换}) \right.$$

右视 (三角代换 / 三角代换)

$$\frac{\sin \sqrt{x}}{\sqrt{x}^3 \sqrt{x}}$$

定积分

黎曼和

微积分基本定理.

$$\int_{a,b} f(x) dx$$

它记 5.1.5.

合部积分 积分上、下限的交换

Wallis 公式

P177 - P178.

周期
奇/偶
 $\times f(\sin x)$

微分 + 积分 $\Rightarrow$ 微分方程.

(解法问题)

(特征)
- 特征非齐次

(§7.2)

二阶常系数微分方程.

特征方程

$$f(x) = x^2$$

$\sinh x, x e^x, \cosh x e^x$

$\lambda = 0$.

$$ax^2 + bx + c$$

积分方程

$\rightarrow$ 完全初始条件.

应用:

稳定性 $\rightarrow$

判断问题.

介值定理 $\rightarrow$

区间上

函数的最大值

函数性质

函数性质

1 单调已证 / 极值

证明局部极值.

6.1-6.3.

{ 平后曲域面积
记数 4 种
弧长 和 曲率

证明:

{ 例 3.3.1.
例 3.1.4.
例 5.2.5